

File Organization

Types of File Organization

1. Serial
 2. Sequential
 3. Random
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Serial File Organization

- Simplest method of file organization
- Data is stored as it comes
- Data is **ALWAYS appended** (added to the end of the file)
- Follows chronological order

Examples: Text files, transaction records, log files

Sequential File Organization

- Data is stored based on **key field sequentially**
- Key fields have to be **unique** AND in **order**

Primary Key	
208	
201	
205	
Key Field	
201	
205	
208	

HITRATE: The number of files that can be accessed at one moment.

- Has **high hitrate** → Can apply batch processing
- Adding new data → Re-arrange memory → Time and power-consuming process

Important: Sequential files **have to be in order**. Inserting data in the middle means moving **ALL data after the insertion point by one position**.

Random File Organization

- Data is stored randomly.
- Records are stored at an address generated by a **hashing algorithm** applied to the key field.

Collision

A collision occurs when one data index is assigned the same memory address as another index.

Reasons for a collision:

1. A record is **already written** at the address given by the hashing algorithm.
2. Two or more records are matched to the **same hash address**.
3. The number of key fields is greater than the number of memory addresses.

Solutions for Collisions

Closed Hashing

- Move line by line to find the first available free space.

Open Hashing (Chaining)

- Also known as **Chaining**
- Create a separate **linked list** of all elements that are in collision.

Overflow Area

- A separate memory area assigned for elements in collision.

HITRATE: Has a low hitrate → Few records can be accessed at a time.

Examples: ATMs, files with usernames and passwords

File Access Methods

1. Sequential Access

- Move element by element to access records.
- Used with **Serial** and **Sequential** file organizations.

2. Direct Access

- Apply a **hashing algorithm** to the key field.
- An address is generated and the system directly accesses the required record.
- **FASTEST method to access data**

If the record is not available, linearly move from the hash address until:

1. Record is found
2. Empty space is found
3. Last address is reached

Used with: Sequential and Random file organizations